DATA ANALYST PORTFOLIO PROJECT

BANK LOAN PERFORMANCE ANALYSIS

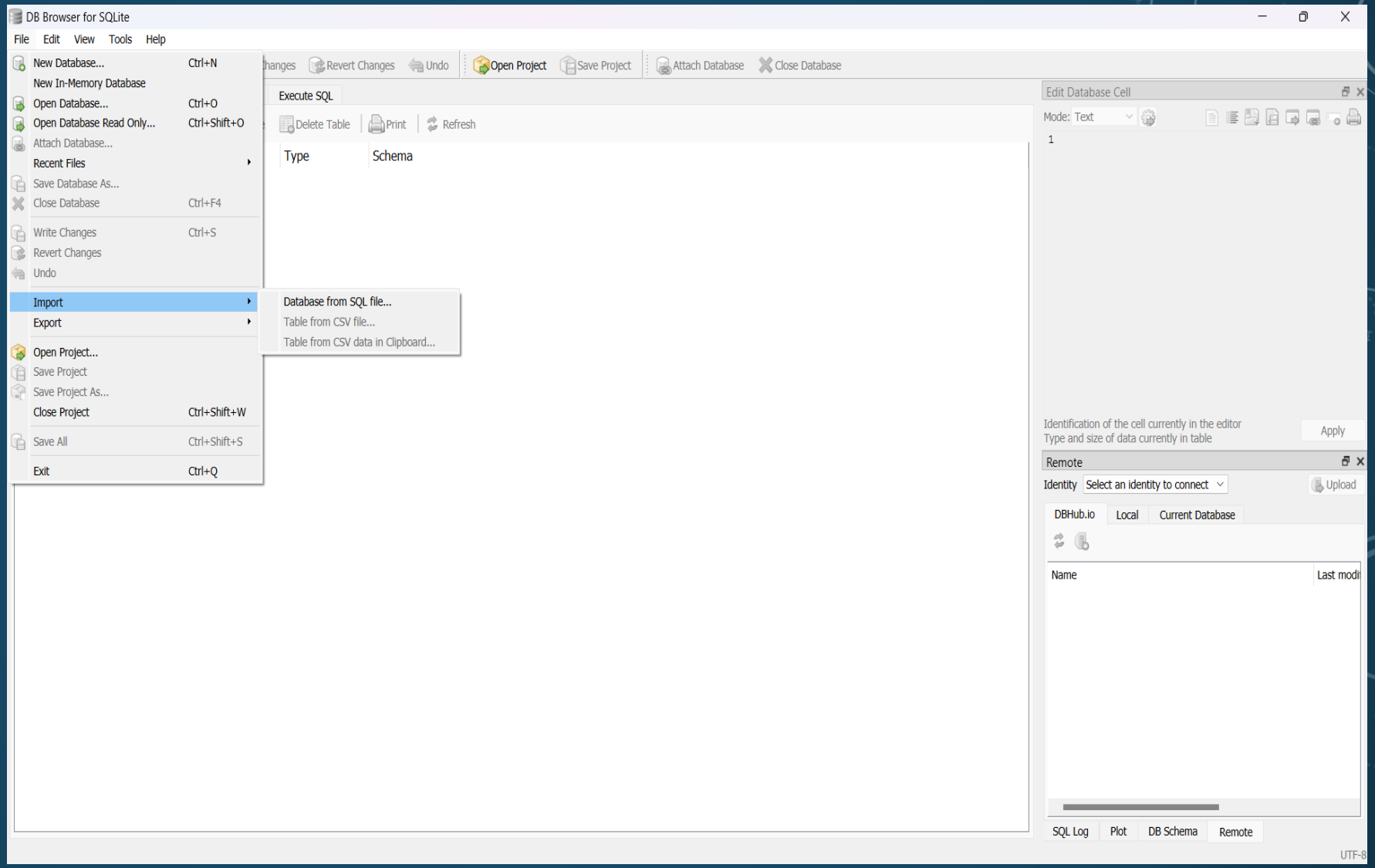
Using SQL and Power BI

BY

Pratiksha L Rohankar

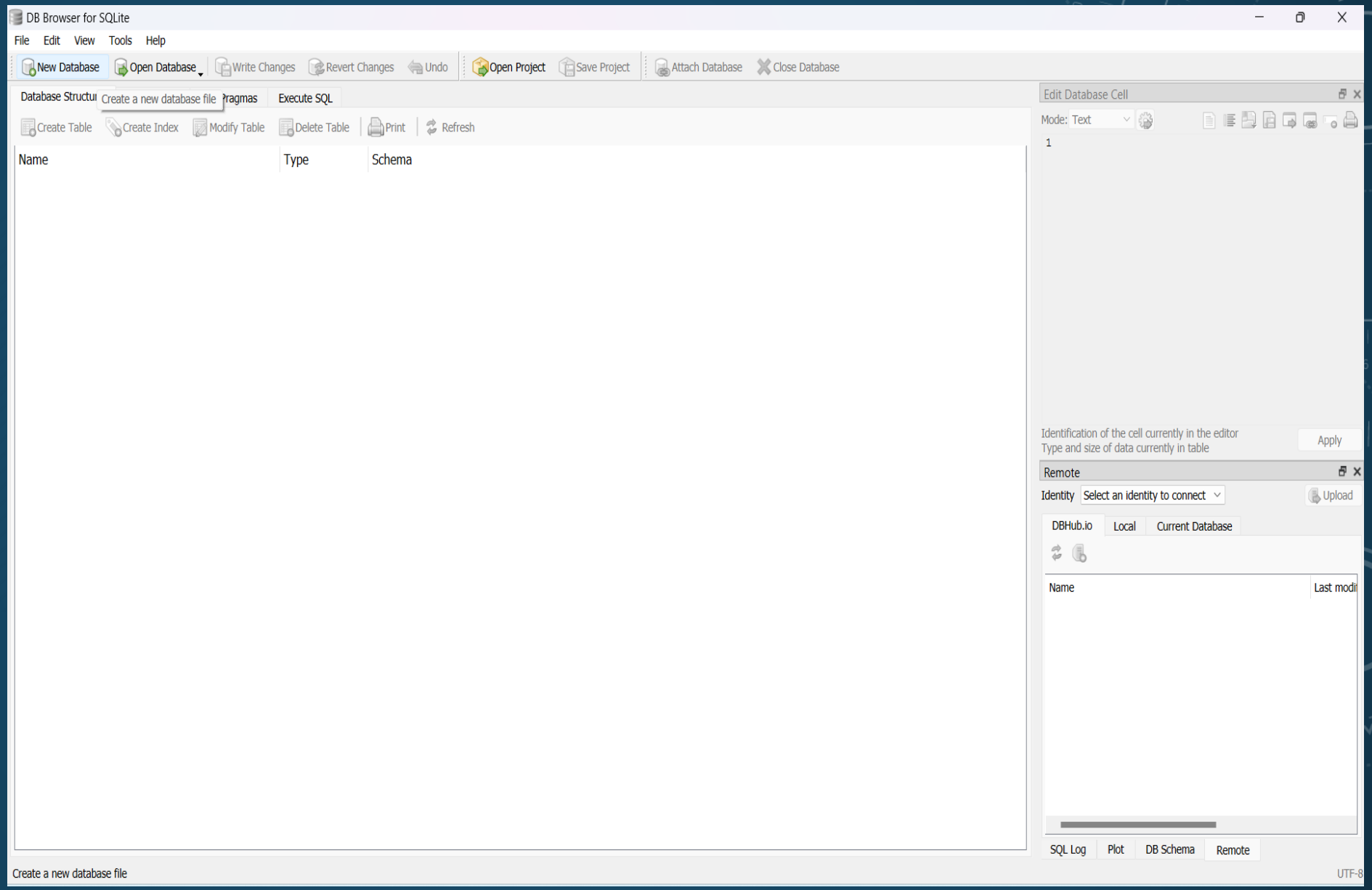
DB Browser for SQLite

IMPORT DATA



DB Browser for SQLite

CREATING DB



DB Browser for SQLite

WRITING QUERIE

DB Browser for SQLite - C:\Users\rohan\Downloads\My Excel Projects\loan_project.sqbpro [loan_project.db]

File Edit View Tools Help

New Database | Open Database | Write Changes | Revert Changes | Undo | Open Project | Save Project | Attach Database | Close Database

Database Structure | Browse Data | Edit Pragmas | Execute SQL

SQL 1

```sql
SELECT * FROM financial_loan;

--Check Total Loans Application
SELECT COUNT(*) AS total_loans
FROM financial_loan;

--Total Funded Amount
SELECT SUM(loan_amount) AS total_funded
FROM financial_loan;

--Total Profit
SELECT SUM(total_payment - loan_amount) AS total_profit
FROM financial_loan;

--Check Average Interest Rate
SELECT
ROUND(AVG(int_rate),4) * 100 AS Avg_Interesr_Rate
FROM financial_loan;

-- Preview Loan Issue Date Records
SELECT issue_date
FROM financial_loan
LIMIT 5;

--December 2021 Average Interest Rate
```

	id	address_state	application_type	emp_length	emp_title	grade	home_ownership	issue_date	last_credit_pull_date	last_payment_d
1	1077430	GA	INDIVIDUAL	< 1 year	Ryder	C	RENT	11-02-2021	13-09-2021	13-04-2021
2	1072053	CA	INDIVIDUAL	9 years	MKC Accounting	E	RENT	01-01-2021	14-12-2021	15-01-2021
3	1069243	CA	INDIVIDUAL	4 years	Chemat Technology Inc	C	RENT	05-01-2021	12-12-2021	09-01-2021
4	1041756	TX	INDIVIDUAL	< 1 year	barnes distribution	B	MORTGAGE	25-02-2021	12-12-2021	12-03-2021
5	1068350	IL	INDIVIDUAL	10+ years	J&J Steel Inc	A	MORTGAGE	01-01-2021	14-12-2021	15-01-2021
6	1062608	CA	INDIVIDUAL	3 years	Studio 94 Corp	C	RENT	17-07-2021	16-03-2021	12-08-2021
7	1067441	TX	INDIVIDUAL	10+ years	American Airlines	C	MORTGAGE	19-11-2021	14-06-2021	13-12-2021
8	1066424	PA	INDIVIDUAL	10+ years	SCI Mahanoy	A	OWN	11-06-2021	14-07-2021	14-07-2021

Edit Database Cell

Mode: Text

1

Editing row=1, column=0
Type: Text / Numeric; Size: 1 character(s)

Apply

Remote

Identity: Select an identity to connect

Upload

DBHub.io | Local | Current Database

Name | Last modi

SQL Log | Plot | DB Schema | Remote

UTF-8

POWER BI

CONNECTING Power BI TO DB Browser for SQLite

POWER BI

RETAIL BANK LOAN PORTFOLIO RISK & PERFORMANCE REPORT | SUMMARY

Summary

Overview

Details

State

All

Grade

All

Purpose

All

Total Loan Applications	Total Funded Amount	Total Received Amount	Avg Interest Rate	Avg DTI
38.6K	$435.8M	$473.1M	12.0%	13.3%
MTD 4.3K	MTD $54.0M	MTD $58.1M	MTD 12.4%	MTD 13.7%
MoM 6.9%	MoM 13.0%	MoM 15.8%	MoM 3.5%	MoM 2.7%

GOOD LOAN ISSUED

86.2%

Good Loan Applications
33.2K

Good Loan Funded Amount
$370.2M

Good Loan Received Amount
$435.8M

BAD LOAN ISSUED

13.8%

Bad Loan Applications
5.3K

Bad Loan Funded Amount
$65.5M

Bad Loan Received Amount
$37.3M

LOAN STATUS

Loan Status	Total Loan Applications	Total Funded Amount	Total Amount Received	MTD Funded Amount	MTD Total Amount Recieved	Avg Interest Rate	Avg DTI
Fully Paid	32145	$35,13,58,350	$41,15,86,256	$4,13,02,025	$4,78,15,851	11.64%	13.17%
Charged Off	5333	$6,55,32,225	$3,72,84,763	$87,32,775	$53,24,211	13.88%	14.00%
Current	1098	$1,88,66,500	$2,41,99,914	$39,46,625	$49,34,318	15.10%	14.72%
Grand Total	38576	$43,57,57,075	$47,30,70,933	$5,39,81,425	$5,80,74,380	12.05%	13.33%

POWER BI

BANK FINANCIAL LOAN REPORT | OVERVIEW

Summary

Overview

Details

Select Measure

Total Amount Receiv...

State

All

Grade

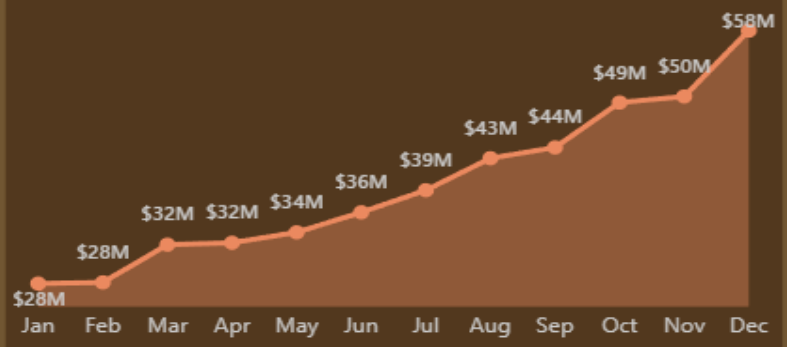
All

Good vs Bad Loan

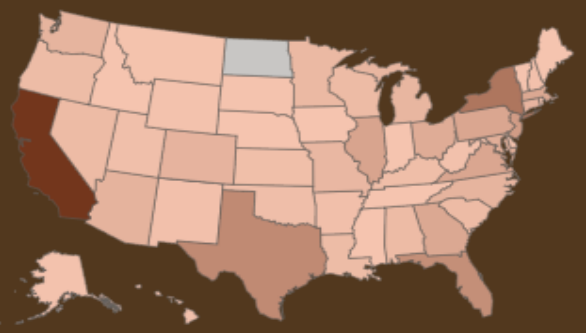
All

Total Loan Applications	Total Funded Amount	Total Received Amount	Avg Interest Rate	Avg DTI
38.6K	$435.8M	$473.1M	12.0%	13.3%
MTD 4.3K	MTD $54.0M	MTD $58.1M	MTD 12.4%	MTD 13.7%
MoM 6.9%	MoM 13.0%	MoM 15.8%	MoM 3.5%	MoM 2.7%

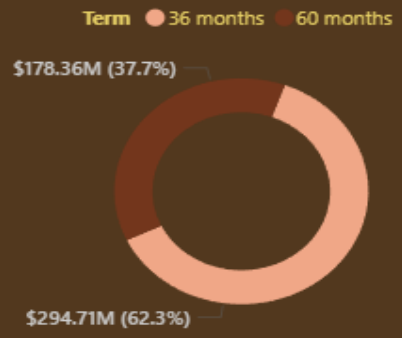
Total Amount Received by Month



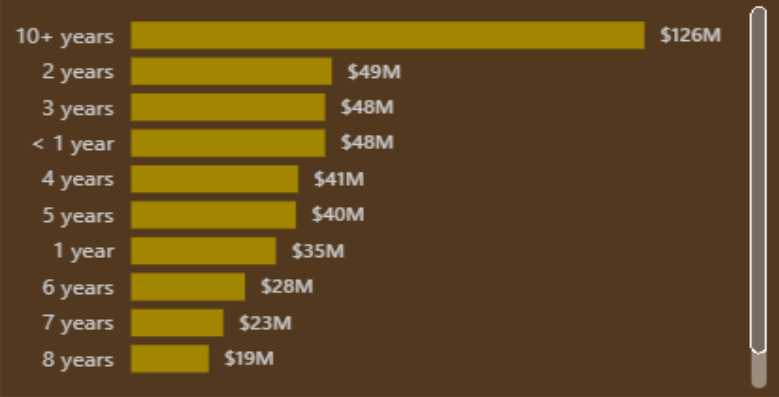
Total Amount Received by State



Total Amount Received by Term

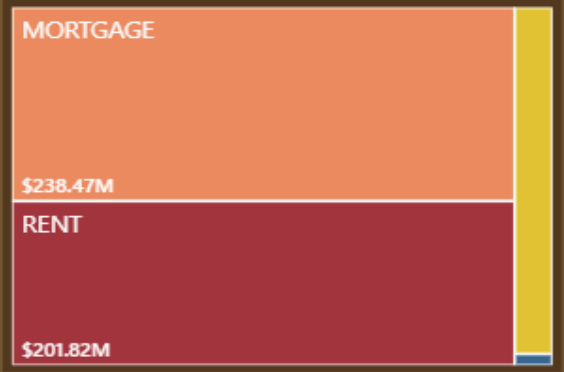


Total Amount Received by Employee Lenght



Total Amount Received by Purpose

Total Amount Received by Home Ownership



POWER BI

BANK FINANCIAL LOAN REPORT | DETAILS

Summary

Overview

Details

Select Measure

Total Amount Receiv...

State

All

Grade

All

Term

All

Total Loan Applications	MTD	MoM
38.6K	4.3K	6.9%

Total Funded Amount	MTD	MoM
$435.8M	$54.0M	13.0%

Total Received Amount	MTD	MoM
$473.1M	$58.1M	15.8%

Avg Interest Rate	MTD	MoM
12.0%	12.4%	3.5%

Avg DTI	MTD	MoM
13.3%	13.7%	2.7%

Id	Purpose	Home Ownership	Grade	Sub Grade	Year	Month	Day	Funded Amount	Int Rate	Installment	Amount Received
779386	other	MORTGAGE	G	G5	2021	June	11	$25,000	0.25	556.03	$32,724
774690	renewable_energy	RENT	G	G4	2021	June	11	$12,000	0.24	348.01	$4,778
976164	Debt consolidation	RENT	G	G5	2021	October	11	$21,250	0.24	612.68	$10,259
1045885	credit card	MORTGAGE	G	G5	2021	December	11	$25,000	0.24	720.80	$31,602
1047612	credit card	MORTGAGE	G	G5	2021	December	11	$24,250	0.24	699.18	$35,627
882097	home improvement	MORTGAGE	G	G4	2021	September	11	$18,000	0.24	516.89	$28,347
968447	Debt consolidation	MORTGAGE	G	G4	2021	October	11	$29,700	0.24	852.86	$37,473
977981	other	MORTGAGE	G	G4	2021	October	11	$17,625	0.24	506.12	$20,896
987992	Debt consolidation	MORTGAGE	G	G4	2021	October	11	$20,000	0.24	574.32	$23,086
998792	Debt consolidation	MORTGAGE	G	G4	2021	November	11	$20,000	0.24	574.32	$31,530
1003497	Debt consolidation	RENT	G	G4	2021	November	11	$25,000	0.24	717.90	$41,470
1034059	Debt consolidation	RENT	G	G4	2021	December	11	$35,000	0.24	1,005.06	$26,028
1040822	Debt consolidation	RENT	G	G4	2021	December	11	$21,575	0.24	619.55	$26,083
1055225	Debt consolidation	RENT	G	G4	2021	December	11	$29,100	0.24	835.63	$40,059
1060644	Debt consolidation	MORTGAGE	G	G4	2021	December	11	$24,250	0.24	519.76	$22,207
1060753	credit card	RENT	G	G4	2021	December	11	$24,925	0.24	715.75	$37,823
747449	major purchase	OWN	G	G5	2021	May	11	$16,000	0.24	456.49	$20,665
763262	Debt consolidation	RENT	G	G5	2021	June	11	$35,000	0.24	851.64	$46,829
831341	wedding	MORTGAGE	G	G5	2021	August	11	$14,500	0.24	413.70	$19,662
842083	Debt consolidation	RENT	G	G5	2021	August	11	$13,650	0.24	389.45	$15,463

PROBLEM STATEMENT

- The bank provides loans to customers across different states, purposes, and terms.
- However, management does not have a clear dashboard to monitor overall loan performance, risk level, and revenue trends.
- The objective of this project is to build an interactive Power BI dashboard to track loan performance, measure risk (Good vs Bad loans), and analyze revenue trends using key performance indicators (KPIs).

DASHBOARD 1: SUMMARY

Key Performance Indicators(KPI's) Requirements

- Total Loan Applications : Track total applications with MTD(Month To Date) and MOM(Month To Month) growth analysis.
- Total Funded Amount : Measure total loan amount gives to customers and monitor monthly performance trends.
- Total Amount Received: Analyze total repayments to evaluate cash flow and recovery efficiency.
- Average Interest Rate : Calculate overall lending rate and track monthly variations.
- Average DTI(Debt-to-Income) : Assess borrower repayment capacity using Debt-to-Income Ratio trends.
- Good vs Bad Loan Ratio : Evaluate portfolio quality by classifying loans based on repayment status.
- Loan Status Distribution : Monitor Fully Paid, Current, and Charged Off loans to understand risk exposure.

DASHBOARD 2: OVERVIEW

Key Performance Indicators(KPI's) Requirements

- Total Amount Received by Month : Analyze monthly revenue trends to identify seasonality patterns and performing growth.
- Total Amount Received by State : Evaluate geographic distribution of loan repayments to identify high-performing regions.
- Total Amount by Loan Term (36 vs 60 Months) : Compare revenue contribution across different loan durations to understand borrower preference and profitability.
- Total Amount by Home Ownership : Assess performance across Mortgage, Rent and Own categories to evaluate risk segmentation.
- Total Amount by Loan Purpose : Identify top revenue-generating loan purposes such as Debt Consolidation, Credit Card, etc.
- Total Amount by Employee Length : Analyze repayment performance based on borrower work experience to evaluate financial stability correlation.

DASHBOARD 3: DETAILS

Key Performance Indicators(KPI's) Requirements

- Loan Identification Metrics : Display Loan ID, Purpose, Grade, Sub-Grade, and Home-Ownership for detailed classification.
- Time Intelligences Breakdown : Include Year, Month, and Day derived from Issue Date to enable advanced time-based analysis.
- Financial Metrics : Show Funded Amount, Installment Amount, Interest Rate, and Amount Received for each loan.
- Filtering and Drill-Down Capability : Enable filtering by State, Grade, Purpose, Term, and Loan Status to allow focused analysis.

THANK YOU